EC-201: Engineering Project Management

LECTURE NO 4: *Project Management Methodologies and Organizational Structures*

Broad Contents

- Project driven versus non-project driven organizations
- Project management methodologies
- Systems, programs, and projects
- Categories of projects
- Product versus Project Management
- Maturity and excellence
- Informal Project Management
- Organizational structures
- Selecting the organizational form

4.1 Project driven versus Non-project driven organizations (1/7)

- On the micro level, virtually all organizations are marketing, engineering, or manufacturing driven. But on the macro level, organizations are either project or non-project driven. In a project driven organization, such as construction or aerospace, all work is characterized through projects, with each project as a separate cost center having its own profit- and-loss statement. The total profit to the corporation is simply the summation of the profits on all projects. In a project driven organization, everything centers on the projects. In the non-project driven organization, such as local or functional manufacturing, profit and loss is measured on vertical technology lines.

4.1 Project driven versus Non-project driven organizations (2/7)

Project management in a non-project driven organization is generally more difficult for these reasons:

- Projects may be few and far between.
- Not all projects have the same project management requirements, and therefore, they cannot be managed identically. This difficulty results from poor understanding of project management and a reluctance of companies to invest in proper training.
- Executives do not have sufficient time to manage projects themselves, yet refuse to delegate authority.
- Projects tend to be delayed because approvals most often follow the vertical chain of command. As a result, project work stays in functional departments.

4.1 Project driven versus Non-project driven organizations (3/7)

- Because project staffing is on a "local" basis, only a portion of the organization understands project management and sees the system in action.
- There exists heavy dependence on subcontractors and outside agencies for project management expertise.

4.1 Project driven versus Non-project driven organizations (4/7)

Non-project driven organizations may also have a steady stream of projects, all of which are usually designed to enhance manufacturing operations. Some projects may be customer requested, such as:

- The introduction of process changes to enhance the final product.
- The introduction of process change concepts to enhance product reliability.

If these changes are not identified as specific projects, the result can be:

- Poorly defined responsibility areas within the organization.
- Slow implementation.
- Lack of a cost tracking system for implementation.

Organization.
communications, both internal and external
to the

4.1 Project driven versus Non-project driven organizations (5/7)

Figure 4.1 below shows the tip-of-the-iceberg syndrome, which can occur in all types of organizations but is most common in non-project driven organizations.

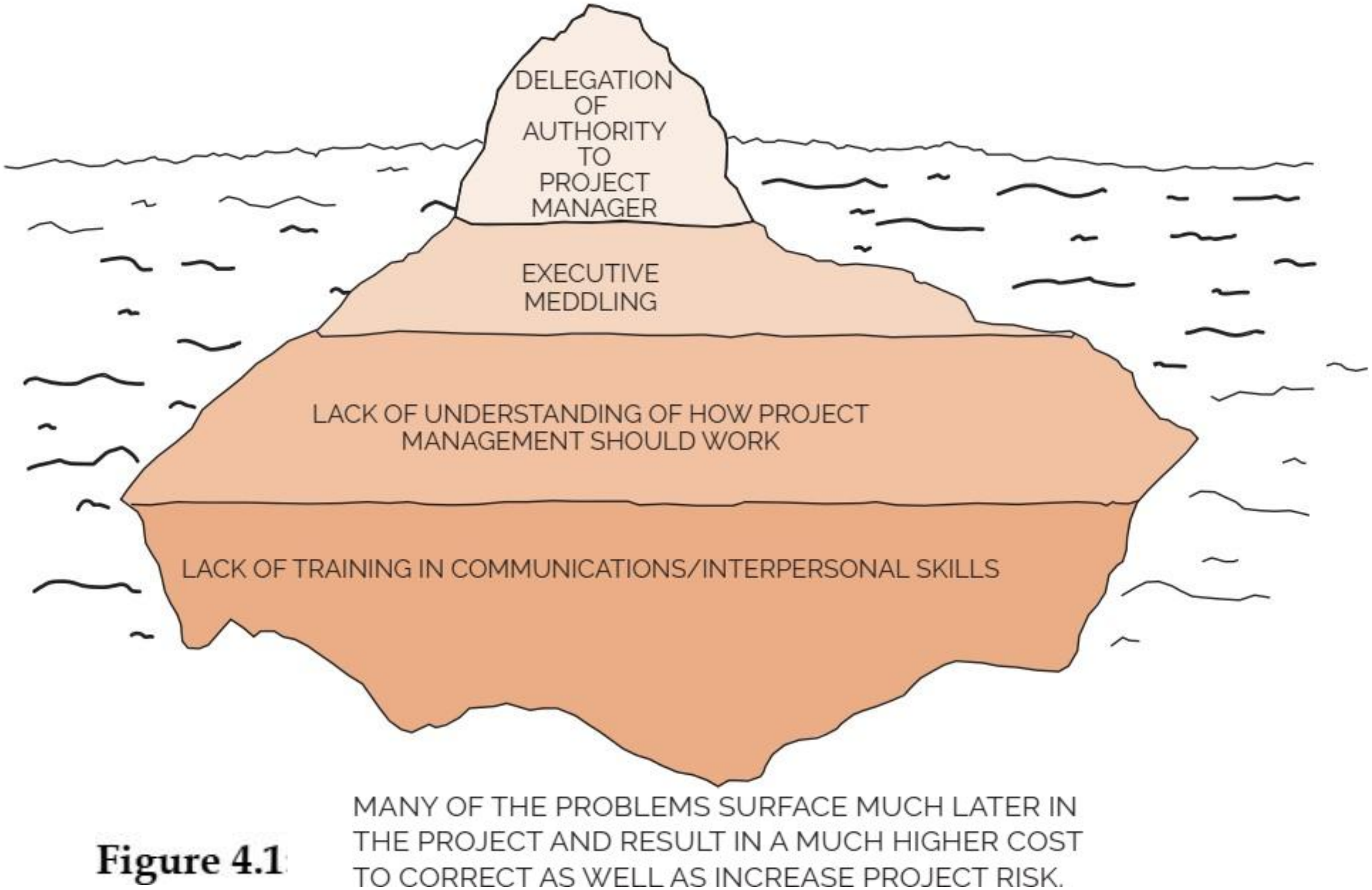


Figure 4.1

4.1 Project driven versus Non-project driven organizations (6/7)

Refer to figure 4.1, previous slide:

- On the surface, all we see is a lack of authority for the project manager. But beneath the surface we see the causes; there is excessive meddling due to lack of understanding of project management, which, in turn, resulted from an inability to recognize the need for proper training.
- In reality, most firms that believed that they were non-project driven were actually hybrids. Hybrid organizations are typically non-project driven firms with one or two divisions that are project driven.
- Historically, hybrids have functioned as though they were non-project driven, as shown in Figure 4.2 next slide, but today they are functioning like project driven firms.

4.1 Project driven versus Non-project driven organizations (7/7)

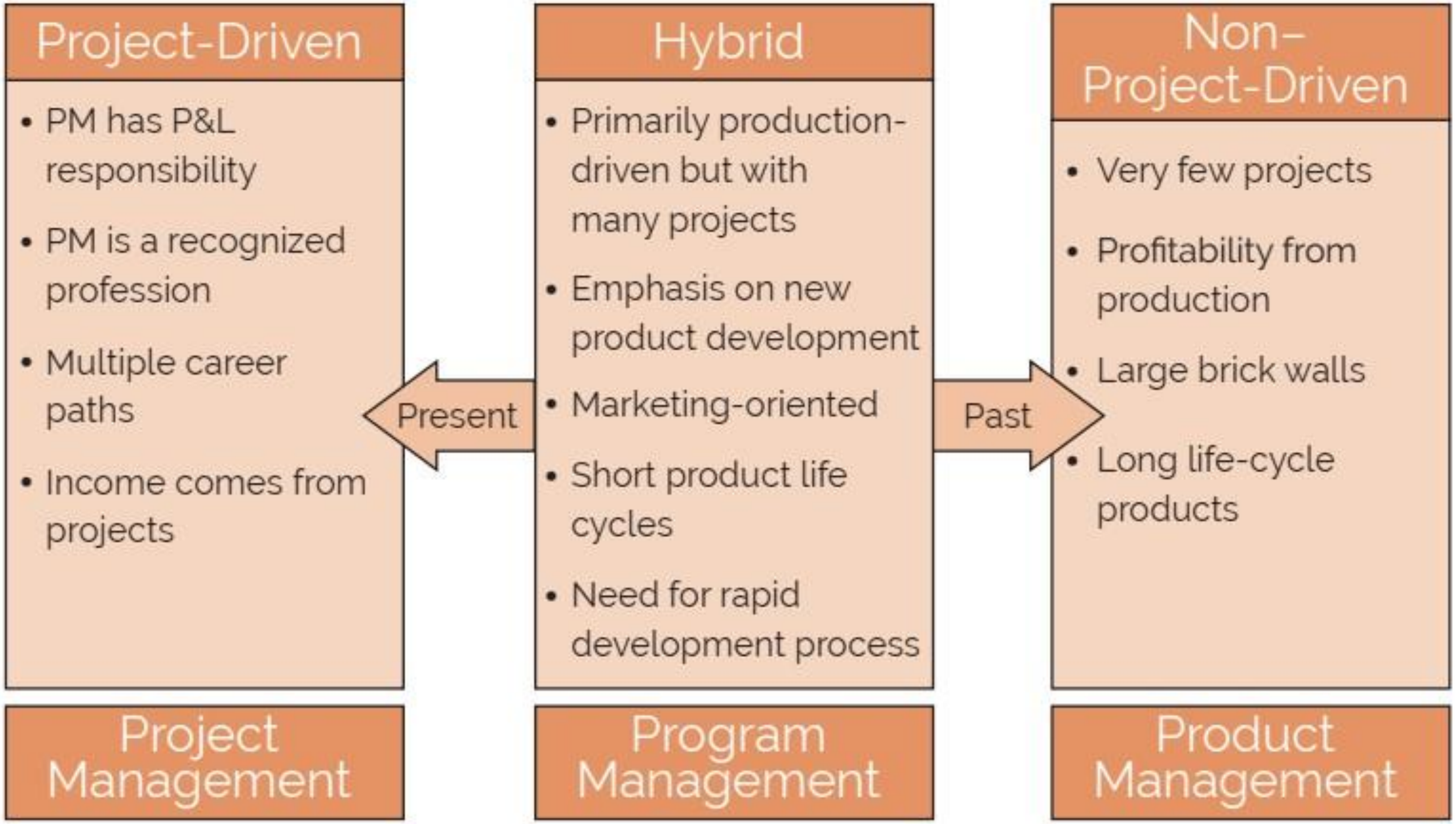


Figure 4.2: Project driven versus non-project driven organizations

4.2 Project Management Methodologies (1/2)

- Achieving project management excellence, or maturity, is more likely with a repetitive process that can be used on each and every project. This repetitive process is referred to as the project management methodology. If possible, companies should maintain and support a single methodology for project management.
- Good methodologies integrate other processes into the project management methodology, as shown in Figure 4.3. Companies have all five of these pro ... ıgement methodology.

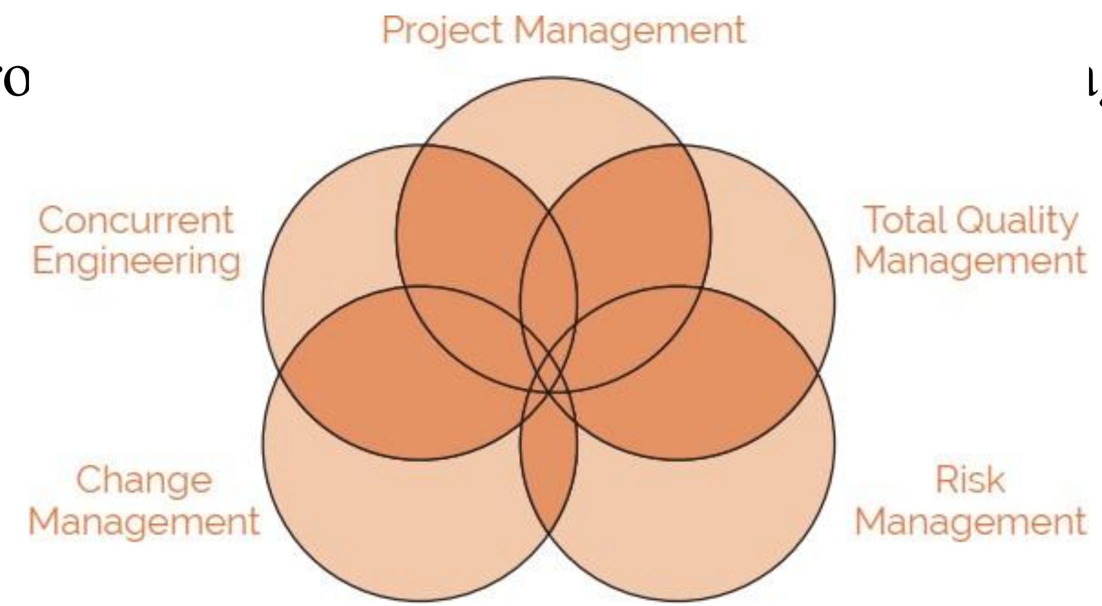


Figure 4.3

4.2 Project Management Methodologies (2/2)

Managing off of a single methodology lowers cost, reduces resource requirements for support, minimizes paperwork, and delimits duplicated efforts. The characteristics of a good methodology based upon integrated processes include:

- A recommended level of detail
- Use of templates
- Standardized planning, scheduling, and cost control techniques
- Standardized reporting format for both in-house and customer use
- Flexibility for application to all projects
- Flexibility for rapid improvements
- Easy for the customer to understand and follow
- Readily accepted and used throughout the entire company
- Use of standardized life-cycle phases and end of phase reviews

4.3 Systems, Programs, and Projects (1/2)

a. **Systems:** *A group of elements, either human or nonhuman, that is organized and arranged in such a way that the elements can act as a whole toward achieving some common goal, objective, or end.*

Systems are characterized by their boundaries or interface conditions. For example, if the business firm system were completely isolated from the environmental system, then a ***close*** system would exist, in which case management would have complete control over all system components. If the business system does in fact react with the environment, then the system is referred to as ***open***. All social systems, for example, are categorized as open systems. Open systems must have reachable boundaries.

4.3 Systems, Programs, and Projects (2/2)

b. ***Programs:*** Programs are generally defined as time-phased efforts, whereas systems exist on a continuous basis.

c. ***Projects:*** Projects are also time-phased efforts (much shorter than programs) and are the first level of breakdown of a program.

4.4 Categories of Projects (1/2)

Once a group of tasks is selected and considered to be a project, the next step is to define the kinds of project units. There are four categories of projects:

1. ***Individual projects:*** These are short-duration projects normally assigned to a single individual who may be acting as both a project manager and a functional manager.
2. ***Staff projects:*** These are projects that can be accomplished by one organizational unit, say a department.
3. ***Special projects:*** Very often special projects occur that require certain primary functions and/or authority to be assigned temporarily to other individuals or units. This works best for short-duration projects. Long-term projects can lead to severe conflicts under this arrangement.

4.4 Categories of Projects (2/2)

4. ***Matrix or Aggregate projects:*** These require input from a large number of functional units and usually control vast resources. Project management is applicable for any ad-hoc (unique, one- time, one-of-a-kind) undertaking concerned with a specific end objective. In order to complete a task, a project manager must:

- Set objectives
- Establish plans
- Organize resources
- Provide staffing
- Set up controls
- Issue directives
- Motivate personnel
- Apply innovation for alternative actions
- Remain flexible

The type of project will often dictate which of these functions a project manager will be required to perform.

4.5 Product versus Project Management

- For all practical purposes, there is no basic difference between product management and project management. Project management and product management are similar, with one major exception: *the project manager focuses on the end date of his project, whereas the product manager is not willing to admit that his product line will ever end*. The product manager wants his product to be as long-lived and profitable as possible.

4.6 Maturity and Excellence

- Some people think that maturity and excellence in project management are the same. Unfortunately, this is not the case. Consider the following definition:
  - *Maturity in project management is the implementation of a standard methodology and accompanying processes, in such a way that ensures a high likelihood of repeated successes.*
- The definition of excellence can be stated as:
  - *Organizations excellence creates an environment in which there exists a continuous stream of successfully managed projects and where success is measured by what is in the best interest of both the company and the project (i.e. the customer).*
- You must have maturity to achieve excellence.

4.7 Informal Project Management

- Informal project management does have some degree of formality but emphasizes managing the project with a minimum amount of paperwork. A reasonable amount of formality still exists. Furthermore, informal project management is based upon guidelines rather than the policies and procedures that are the basis for formal project management. Informal project management mandates:
  - Effective communications
  - Effective cooperation
  - Effective teamwork
  - Trust
- These four elements are absolutely essential for informal project management to work effectively.

4.8 Organizational Structures (1/13)

- Management has come to realize that organizations must be dynamic in nature; that is, they must be capable of rapid restructuring, if environmental conditions so dictate. These environmental factors evolved from the increasing competitiveness of the market, changes in technology, and a requirement for better control of resources for multiproduct firms.
- Much has been written about how to identify and interpret those signs that indicate that a new organizational form may be necessary. According to Grinnell and Apple, there are five general indications that the traditional structure may not be adequate for managing projects:
    - Management is satisfied with its technical skills, but projects are not meeting time, cost, and other project requirements.
    - There is a high commitment to getting project work done, but gre a t

4.8 Organizational Structures (2/13)

- Highly talented specialists involved in the project feel exploited and misused.
- Particular technical groups or individuals constantly blame each other for failure to meet specifications or delivery dates.
- Projects are on time and to specifications, but groups and individuals are not satisfied with the achievement.

Unfortunately, many companies do not realize the necessity for organizational change until it is too late. For each of the organizational structures advantages and disadvantages are listed. Many of the disadvantages stem from possible conflicts arising from problems in authority, responsibility, and accountability.

4.8 Organizational Structures (3/13)

1. ***Traditional (Classical) Organization:*** The traditional management structure has survived for more than two centuries. Fifty years ago companies could survive with only one or perhaps two product lines. The classical management organization is shown in figure 4.4 next slide. The classical management organization was found to be satisfactory for control, and conflicts were at a minimum. However, with the passing of time, companies found that survival depended on multiple product lines (that is diversification) and vigorous integration of technology into the existing organization.

4.8 Organizational Structures (4/12)

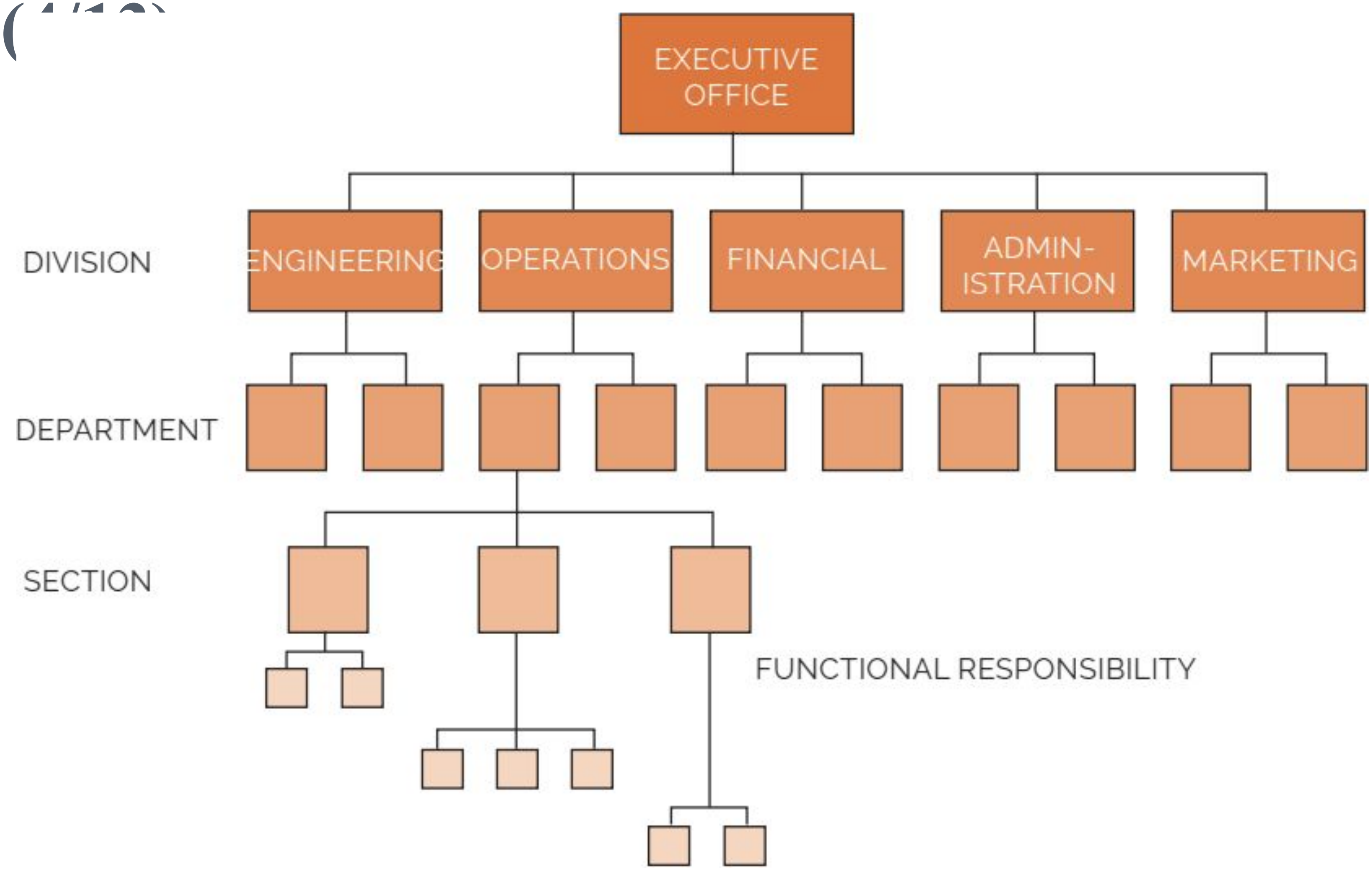


Figure 4.4

4.8 Organizational Structures (5/13)

1. ***Traditional (Classical) Organization:*** The most common advantages and disadvantages of this type of organizations are listed in tables 4.1 and 4.2 respectively.

TABLE 4–1. ADVANTAGES OF THE TRADITIONAL (CLASSICAL) ORGANIZATION

- Easier budgeting and cost control are possible.
- Better technical control is possible.
- Flexibility in the use of manpower.
- Continuity in the functional disciplines; policies, procedures, and lines of responsibility are easily defined and understandable.
- Quick reaction capability exists, but may be dependent upon the priorities of the functional managers.

TABLE 4–2. DISADVANTAGES OF THE TRADITIONAL (CLASSICAL ORGANIZATION)

- No one individual is directly responsible for the total project (i.e., no formal authority; committee solutions).
- Coordination becomes complex, and additional lead time is required for approval of decisions.
- Decisions normally favor the strongest functional groups.
- Response to customer needs is slow.
- Motivation and innovation are decreased.

4.8 Organizational Structures (6/13)

2. ***Pure Product (Projectized) Organization:*** The pure product organization develops as a division within a division. The projectized organization is shown in figure 4.5 next slide. As long as there exists a continuous flow of projects, work is stable and conflicts are at a minimum. The major advantage of this organizational flow is that one individual, the program manager, maintains complete line authority over the entire project. Not only does he assign work, but he also conducts merit reviews. Because each individual reports to only one person, strong communication channels develop that result in a very rapid reaction time. Functional managers were able to maintain qualified staffs for new product development without sharing personnel with other programs and projects. Upper-level management was now able to spend more time on executive decision making than on confli c t arbitration.

4.8 Organizational Structures (7/13)

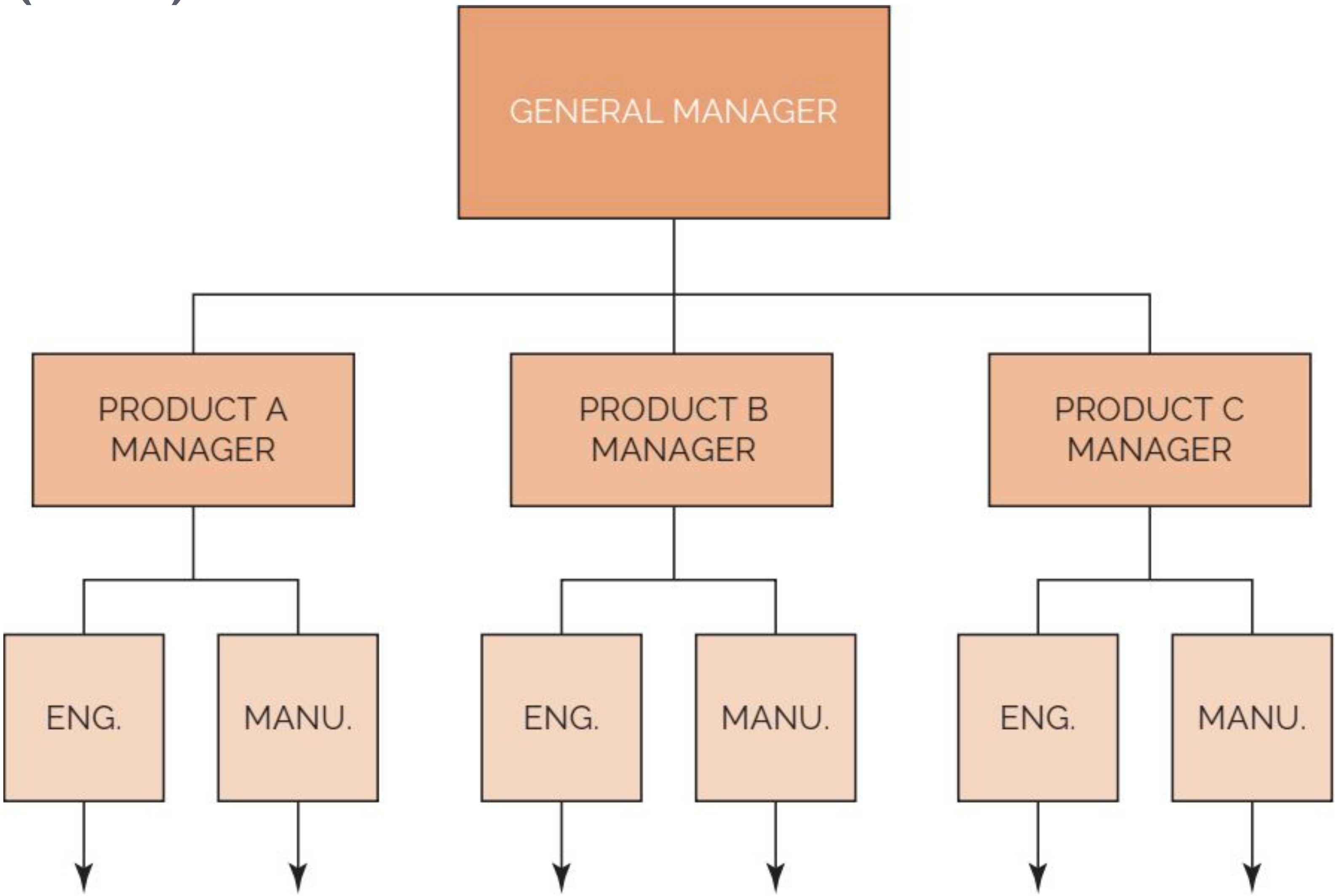


Figure 4.5

4.8 Organizational Structures (8/13)

2. ***Pure Product (Projectized) Organization:*** The most common advantages and disadvantages of this type of organizations are listed in tables 4.3.

TABLE 4–3. ADVANTAGES AND DISADVANTAGES OF THE PRODUCT ORGANIZATIONAL FORM

Advantages

- Provides complete line authority over the project (i.e., strong control through a single project authority).
- Participants work directly for the project manager. Unprofitable product lines are easily identified and can be eliminated.
- Strong communications channels.
- Very rapid reaction time is provided.
- Upper-level management maintains more free time for executive decision making.

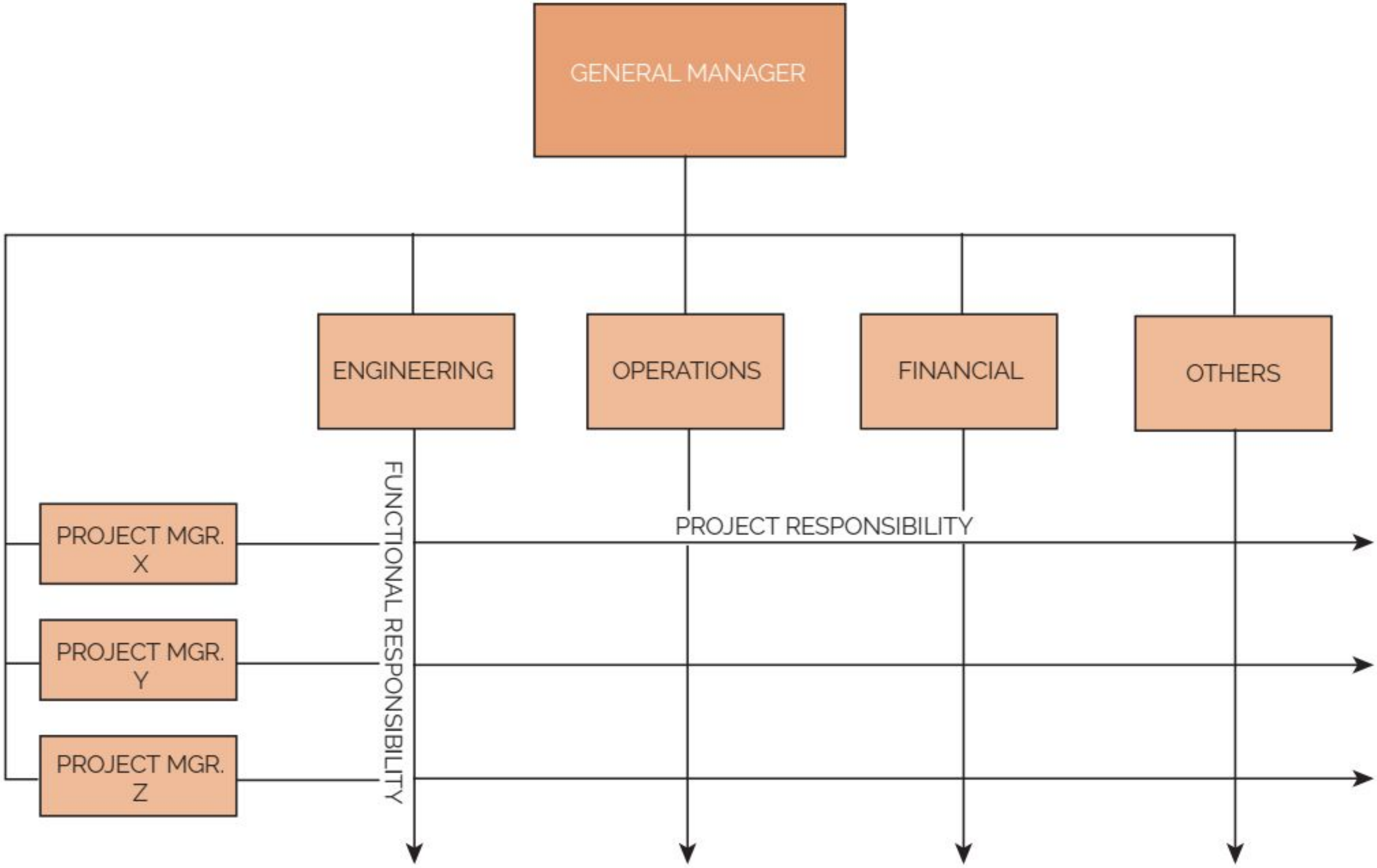
Disadvantages

- A tendency to retain personnel on a project long after they are needed. Upper-level management must balance workloads as projects start up and are phased out.
- Technology suffers because, without strong functional groups, outlook of the future to improve company's capabilities for new programs would be hampered (i.e., no perpetuation of technology).
- Control of functional (i.e., organizational) specialists requires top-level coordination.
- Lack of opportunities for technical interchange between projects.
- Lack of career continuity and opportunities for project personnel.

4.8 Organizational Structures (9/13)

3. Matrix Organizational Form: The matrix organizational form is an attempt to combine the advantages of the traditional structure and the product organizational structure. This form is ideally suited for companies, such as construction, that are "project-driven." The projectized organization is shown in figure 4.6 next slide. Each project manager reports directly to the vice president and general manager. The project manager has total responsibility and accountability for project success. Each functional unit is headed by a department manager whose prime responsibility is to ensure that a unified technical base is maintained and that all available information can be exchanged for each project. Department managers must also keep their people aware o f the latest technical accomplishments

4.8 Organizational Structures (10/13)



4.8 Organizational Structures (11/13)

3. Matrix Organizational Form:

Certain ground rules exist for matrix development. These are:

- Participants must spend full time on the project; this ensures a degree of loyalty.
- Horizontal as well as vertical channels must exist for making commitments.
- There must be quick and effective methods for conflict resolution.
- There must be good communication channels and free access between managers.
- All managers must have input into the planning process.
- Both horizontally and vertically oriented managers must be willing to negotiate for resources.
- The horizontal line must be permitted to operate as a separate entity except for administrative purposes.

4.8 Organizational Structures (12/13)

3. *Matrix Organizational Form:*

Each ground rule brings with it common advantages and disadvantages that are described in tables 4.4 and 4.5 respectively

TABLE 4–4. ADVANTAGES OF A PURE MATRIX ORGANIZATIONAL FORM

Advantages

- The project manager maintains maximum project control (through the line managers) over all resources, including cost and personnel.
- Policies and procedures can be set up independently for each project, provided that they do not contradict company policies and procedures.
- The project manager has the authority to commit company resources, provided that scheduling does not cause conflicts with other projects.
- The functional organizations exist primarily as support for the project.
- Because key people can be shared, the program cost is minimized. People can work on a variety of problems; that is, better people control is possible.
- Conflicts are minimal, and those requiring hierarchical referrals are more easily resolved.
- There is a better balance among time, cost, and performance.
- Authority and responsibility are shared.

4.8 Organizational Structures (13/13)

3. Matrix Organizational Form:

TABLE 4–5. DISADVANTAGES OF A PURE MATRIX ORGANIZATIONAL FORM

Disadvantages
• Multidimensional information flow. • Dual reporting. • Potential for continuous conflict and conflict resolution. • More effort and time are needed initially to define policies and procedures, compared to traditional form. • Balance of power between functional and project organizations must be watched. • Balance of time, cost, and performance must be monitored. • Conflicts and their resolution may be a continuous process (possibly requiring support of an organizational development specialist). • People do not feel that they have any control over their own destiny when continuously reporting to multiple managers.

4.9 Selecting the Organizational Form (1/3)

- Project management has matured as an outgrowth of the need to develop and produce complex and/or large projects in the shortest possible time, within anticipated cost, with required reliability and performance, and (when applicable) to realize a profit.
- Modern organizations have become so complex that traditional organizational structures and relationships no longer allow for effective management, how can executives determine which organizational form is best, especially since some projects last for only a few weeks or months while others may take years?

4.9 Selecting the Organizational Form (2/3)

- To answer such a question, we must first determine whether the necessary characteristics exist to warrant a project management organizational form. Generally speaking, the project management approach can be effectively applied to a one-time undertaking that is:
  - Definable in terms of a specific goal
  - Infrequent, unique, or unfamiliar to the present organization
  - Complex with respect to interdependence of detailed tasks
  - Critical to the company

4.9 Selecting the Organizational Form (3/3)

- The basic factors that influence the selection of a project organizational form are:
  - Project size (small scale, medium scale and large scale project)
  - Project length (Project start and end dates)
  - Experience with project management organization
  - Philosophy and visibility of upper-level management
  - Project location
  - Available resources
  - Unique aspects of the project

References:

† *Harold Kerzner, "Project Management, A System Approach to Planning, Scheduling & Controlling", 12th Edition, Wiley.*

† *James. P. Lewis "Project Planning, Scheduling, and Control", 3rd Edition, McGraw-Hill Companies.*

† *TimothyJ. Kloppenborg, Vittal Anantatmula, Kathryn N. Wells, "Contemporary Project Management" 4th Edition, Cengage Learning.*